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NPS/005/001 – Technical Specification for Teleprotection Intertripping Equipment

1. Purpose

The purpose of this document is to detail the requirements for Teleprotection Intertripping Equipment for use in primary substations and protection schemes of the CE-Electric UK distribution network.

2. Scope

This document applies to the design specification, operational requirements and construction for Teleprotection Intertripping equipment.

It is preferred that the Teleprotection Intertripping equipment unit is approved by the ENA Protection Approval Panel to ENATS 48-5. CE-Electric UK reserves the right to complete self approval on other units.

The equipment shall be in compliance with the technical standards detailed in this document.

3. Equipment Requirements

3.1 Channels

3.1.1 Each unit shall be capable of providing a single or two independent Intertripping channels.

3.1.2 The device shall be full duplex to allow transmission in both directions simultaneously.

3.1.3 Each intertripping channel shall be capable of providing an intertrip output at the receiving end within the following specified times from reception of the incoming trip at the sending end:

Standard Speed	100 mSeconds	minimum requirement
High Speed	50 mSeconds	minimum requirement
Very High Speed	25 mSeconds	minimum requirement
	15 mSeconds	preferred requirement

These operating speeds include the operation of the output relay.

3.1.4 Each intertripping channel input shall be capable of being triggered from an independent command input of either:

24V dc (nominal)
48 V dc (nominal)
110V dc (nominal)

The command input voltage shall comply with ENATS 48-5.

3.1.5 The security and dependability of each intertripping channel shall comply with EN 60834-1.

3.1.6 Each Intertripping channel output shall be capable of providing a minimum of two volts free contacts, normally open, closed on receipt of a trip. Output contacts shall be capable of driving a high burden tripping relay and comply with ENATS 48-4.

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3.2 Environmental Requirements

- 3.2.1 Atmospheric -. Each unit shall complying with ENATS 48-5 section 4.
- 3.2.2 Mechanical -. Each unit shall complying with ENATS 48-5 section 5.
- 3.2.3 Electrical -. Each unit shall complying with ENATS 48-5 section 6.
- 3.2.4 Electromagnetic Compatibility -. Each unit shall complying with ENATS 48-5 section 7.

3.3 Communications

- 3.3.1 The unit shall have a common signalling output channel which BABT approved.
- 3.3.2 The unit shall have the ability to support analogue 4 – wire communication links, one pair TX and one pair RX.
- 3.3.3 The unit shall have communications line terminations that are matched for 600 Ohm impedances on both the transmitter and receiver.
- 3.3.4 The unit shall have the ability to vary the transmitted signal level in 1dB steps from -13dBm (maximum) to -28dBm (minimum).
- 3.3.5 The unit shall have the ability to operate with a receive signal at a level in the range from -16dBm (maximum) to -30dBm (minimum).
- 3.3.6 It is preferred that the unit has the option of connecting to a digital communications interface (X21 interface).

3.4 Alarms

- 3.4.1 The unit shall provide two alarm output volt free contacts, normally open, closed on failure of the unit.
- 3.4.2 The unit shall fail on detection of faulty hardware, receipt of corrupted data, loss of incoming data/carrier signal, failure of incoming power supply or supply voltage out of limits.

3.5 Supplies

- 3.5.1 The unit shall be able to be powered from the following supply voltages:

- 24V dc (nominal)
- 48V dc (nominal)
- 110V dc (nominal)
- The supply voltage shall comply with ENATS 48-5.

- 3.5.2 The unit shall provide a method of safely removing the supply voltage for isolation purposes.

3.6 Interface

- 3.6.1 Where the unit is to replace an existing unit then it is preferred that the unit has the option to inter-network with existing Teleprotection signalling protocols.

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3.7 Mechanical

3.7.1 The unit shall be capable of being mounted in standard 19 inch cubicle rack mounts. Cable looms shall be provided to allow interconnections to be made at the rear of the cubicle.

3.7.2 All component cards shall be housed behind removable facia plates or lockable door panels.

4. References

4.1 External Documentation

Reference	Title
ENATS 48-5	ENA Technical Specification Protection Approval Panel Environmental Tests for Protection Relays and Systems
ENATS 48-4	ENA Technical Specification DC Relays Associated with a Tripping Function in Protection Systems
EN 60834-1	Teleprotection Equipment of Power Systems Performance and Testing Part 1: Command Systems
BABT Approvals CEGB Standards and General Specification for Electronic Equipment P21	EES1980 & EES1989 High Speed Voice Frequency Intertripping Equipment

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Ian D Aynsley	Project Coordinator	I D Aynsley	22/01/2007

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Geoff Bolden	Purchasing Manager	Geoff Bolden	23/01/2007

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

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Yvonne Dobbison	Technical Services Engineer	Yvonne Dobbison	22/01/2007

6.4 DBD Assurance

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Sean Johnson	DBD Administrator	Sean Johnson	05/02/2007

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Jim Paine	Technical Services Manager	Jim Paine	23/01/2007

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Mike Leighton	Director of Asset Construction	Mike Leighton	01/02/2007